

Smart Company CareerMatch

AI-Powered Job Search, Resume Matching & Career Preparation Assistant

A unified Streamlit application that connects job discovery, resume analysis, skill matching and role-specific interview preparation.

JOB SEARCH

RESUME ANALYSIS

JOB MATCHING

INTERVIEW PREP

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Problem Statement

Challenges faced during job discovery and interview preparation

Scattered job discovery

Opportunities are spread across company career pages and public ATS platforms. Users often repeat the same search.

Resume–job mismatch

Candidates need a quick way to compare their skills with a specific role and identify missing skills.

Generic preparation

Interview preparation is often disconnected from the selected role. Focused, role-aware questions are more useful.

Project objective: connect discovery → analysis → matching → preparation in one workflow.

Proposed Solution

A single application for the complete career-preparation journey

Search

01

Natural-language
job/company search

Analyse

02

Extract skills from a resume

Match

03

Compare candidate and job
skills

Prepare

04

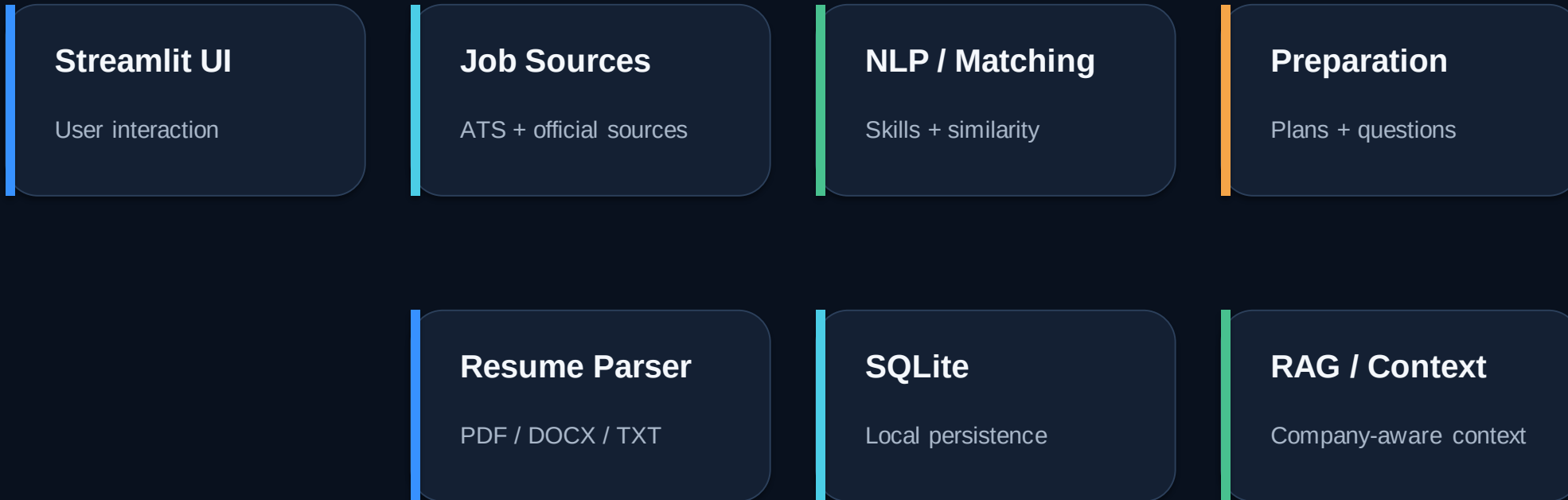
Create preparation and
interview questions

Core value: reduce the gap between finding a job and knowing how to prepare for it.

Fast focused search • optional broader live-source search • resume parsing • skill matching • skill-gap identification • personalised preparation • 100 interview questions

System Architecture

Modular components and information flow

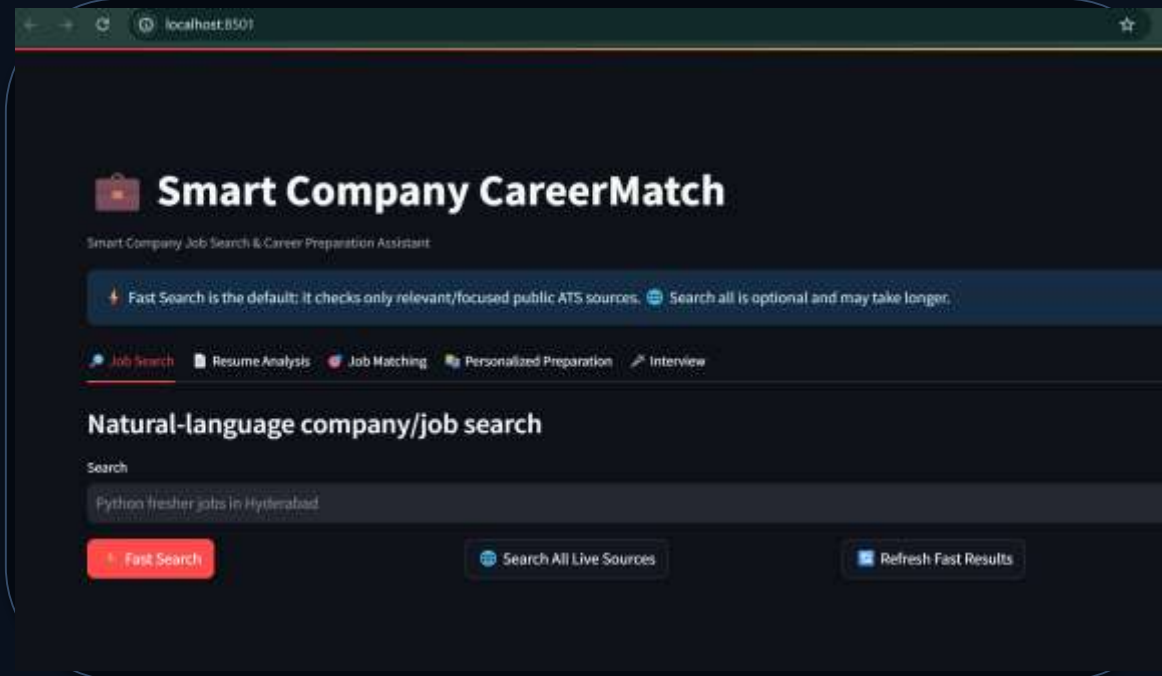


Python • Streamlit • SQLite • NLP/skill matching • public job-source connectors

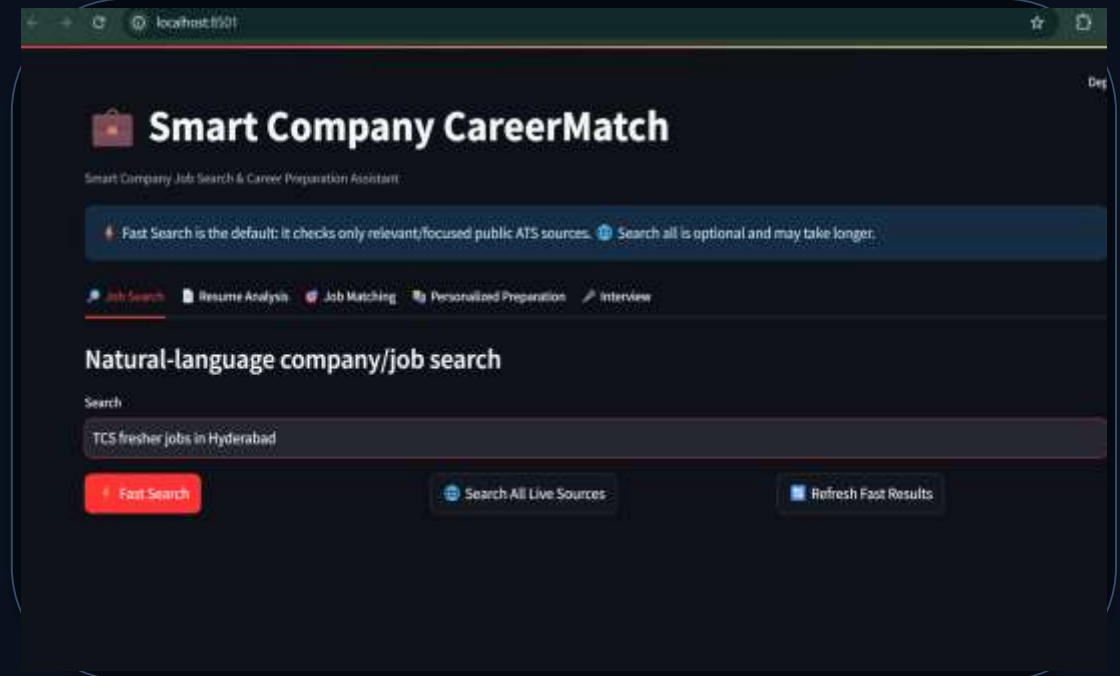
Separation of modules keeps retrieval, parsing, matching and preparation logic easier to test and extend.

Feature 01 — Natural-Language Job Search

Users can search with everyday phrases such as “TCS fresher jobs in Hyderabad”.



Application home and search interface

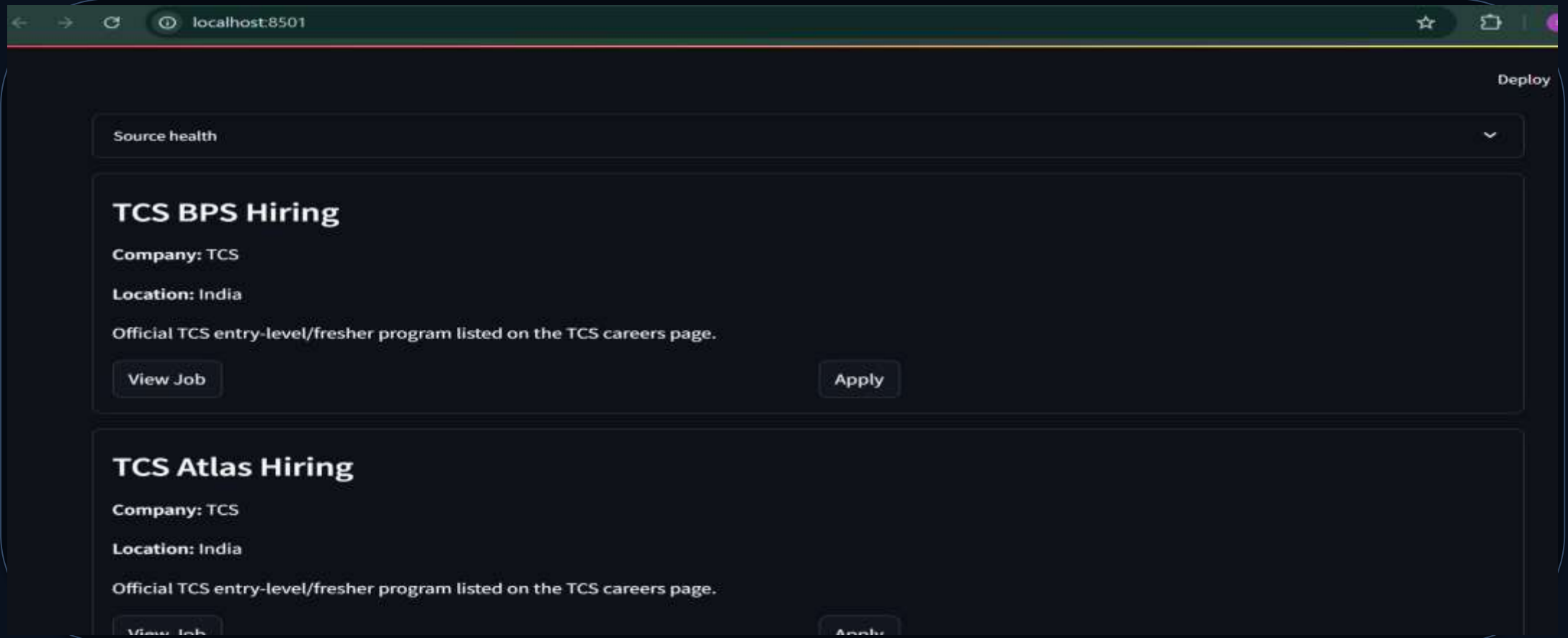


Example: TCS fresher job search

The interface supports fast focused search and an optional broader live-source search, helping users discover relevant roles with less manual effort.

Job Discovery — Source Results

Structured result cards make job opportunities easy to scan and act on.

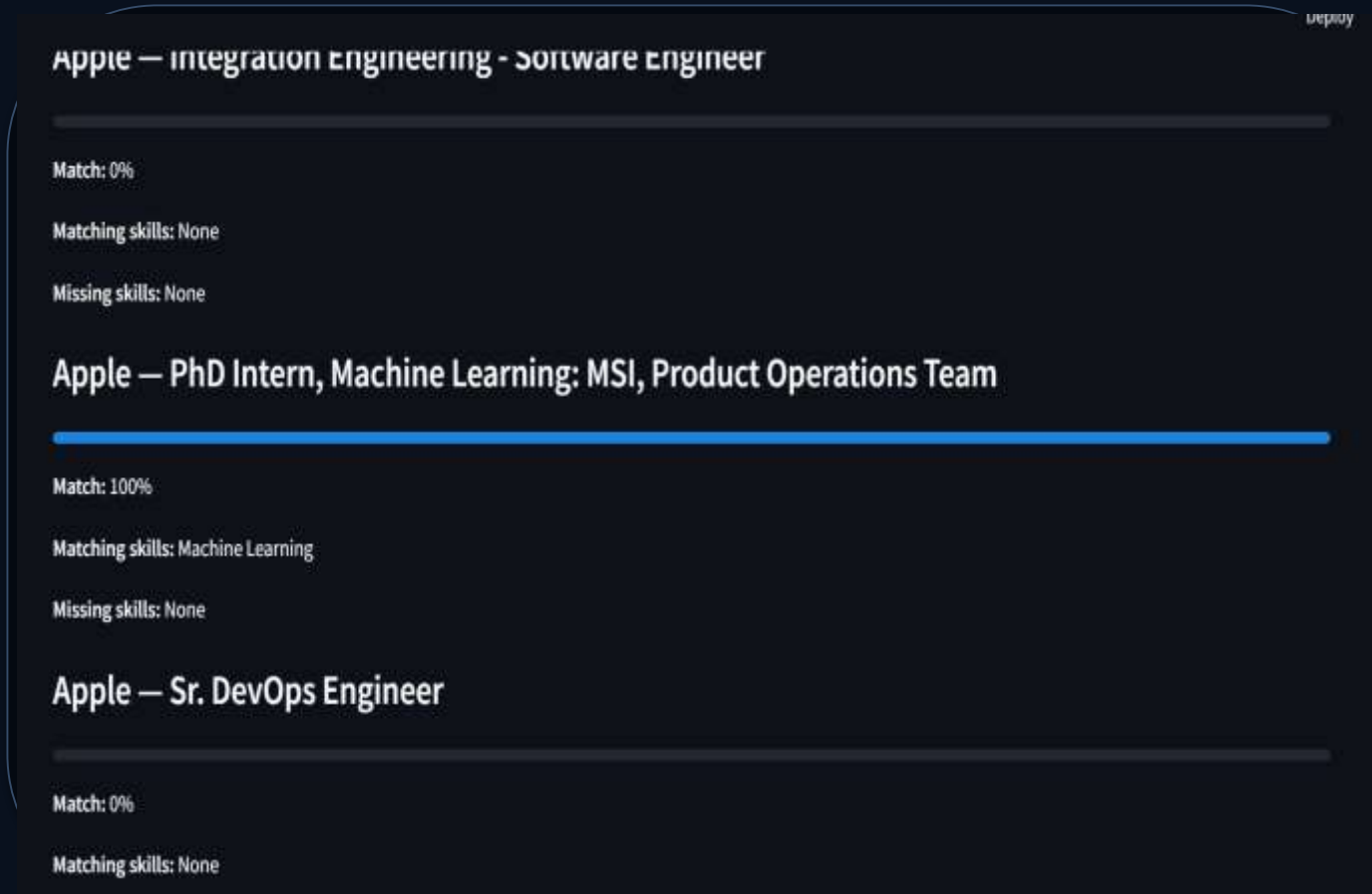


Example TCS results with company, location, role context and actions

View Job and Apply actions connect the discovery experience to the relevant opportunity.

Feature 02 — Resume-to-Job Skill Matching

The matching module compares extracted candidate skills with the selected role.



Example matching results

Match percentage

A simple percentage communicates the degree of skill alignment.

Matching skills

Shows skills found in both the candidate profile and the job.

Missing skills

Highlights gaps that can guide learning and preparation.

Feature 03 — Role-Specific Interview Preparation

Generates a large, role-aware set of questions for the selected job.

Job Search

Resume Analysis

Job Matching

Personalized Preparation

Interview

100 Relevant Interview Questions

Select a job

Apple — IN-Business Expert

1. [Apple — Retrieved] How would you debug a performance issue in a user-facing application?

2. [Apple — Retrieved] How would you design a reliable feature with a strong user experience?

3. [Apple — Retrieved] How do you think about memory, concurrency, and resource management?

4. [Apple — Retrieved] How would you test software across different devices or operating conditions?

5. [Apple — Retrieved] Describe a project where attention to detail materially improved the result.

6. [Apple — Retrieved] How would you investigate a crash that occurs only for some users?

7. [Apple — Retrieved] How would you balance simplicity, performance, privacy, and maintainability?

8. [Apple — Retrieved] How would you collaborate with designers, hardware teams, and software engineers?

localhost:8501

79. [Job-specific] For the IN-Business Expert role, interview problem 45: explain your assumptions, solution, testing strategy, and trade-offs.

80. [Job-specific] For the IN-Business Expert role, interview problem 46: explain your assumptions, solution, testing strategy, and trade-offs.

81. [Job-specific] For the IN-Business Expert role, interview problem 47: explain your assumptions, solution, testing strategy, and trade-offs.

82. [Job-specific] For the IN-Business Expert role, interview problem 48: explain your assumptions, solution, testing strategy, and trade-offs.

83. [Job-specific] For the IN-Business Expert role, interview problem 49: explain your assumptions, solution, testing strategy, and trade-offs.

84. [Job-specific] For the IN-Business Expert role, interview problem 50: explain your assumptions, solution, testing strategy, and trade-offs.

85. [Job-specific] For the IN-Business Expert role, interview problem 51: explain your assumptions, solution, testing strategy, and trade-offs.

86. [Job-specific] For the IN-Business Expert role, interview problem 52: explain your assumptions, solution, testing strategy, and trade-offs.

87. [Job-specific] For the IN-Business Expert role, interview problem 53: explain your assumptions, solution, testing strategy, and trade-offs.

88. [Job-specific] For the IN-Business Expert role, interview problem 54: explain your assumptions, solution, testing strategy, and trade-offs.

89. [Job-specific] For the IN-Business Expert role, interview problem 55: explain your assumptions, solution, testing strategy, and trade-offs.

90. [Job-specific] For the IN-Business Expert role, interview problem 56: explain your assumptions, solution, testing strategy, and trade-offs.

91. [Job-specific] For the IN-Business Expert role, interview problem 57: explain your assumptions, solution, testing strategy, and trade-offs.

Interview tab — first question set

Interview tab — job-specific question set

Questions include retrieved role-oriented prompts and job-specific interview problems.

Technology Stack

Technologies used to build the current prototype

Python

Core application and business logic

Streamlit

Interactive web interface

SQLite

Local database and persistence

NLP / Skill Matching

Resume and job comparison

PDF / DOCX / TXT

Resume ingestion

Public ATS Sources

Greenhouse, Lever, Ashby and official sources

Modular Python files • automated tests • environment configuration via .env

End-to-End User Workflow

From discovering an opportunity to preparing for its interview



Project Outcomes & Future Enhancements

Current capabilities and possible next steps

Current capabilities

- Natural-language job/company search
- Public-source / ATS retrieval
- Resume parsing and skill extraction
- Resume-to-job matching
- Skill-gap identification
- Personalised preparation
- 100 relevant interview questions

Future enhancements

- More official career integrations
- Stronger semantic matching
- Duplicate detection and ranking controls
- Richer resume feedback
- Authentication and cloud deployment
- Job alerts and saved searches
- Interview answer evaluation

Vision: evolve from a job-search utility into a complete, explainable career-preparation platform.

Thank You

Smart Company CareerMatch

A practical Python + Streamlit project connecting job discovery, resume intelligence, skill matching and interview preparation.

Questions & Discussion

Shaik Mahammad Abdul Zaib